

Practice Tests 1

Course Name: Database Principle Examination Time: 120minutes

Student Number: _____ Student Name: _____

1. What is database integrity? How is it different from database security? (5+5)
2. What is the basic principle of database recovery? What is the implementation of database recovery? (4 + 6)
3. The relations R and S are shown in the figure below. Please provide the results of the following expressions. (5+10)

R				S	
A	B	C	D	C	D
a ₁	b ₁	c ₁	d ₁	c ₁	d ₁
a ₁	b ₁	c ₂	d ₂	c ₂	d ₂
a ₁	b ₁	c ₁	d ₃		
a ₂	b ₂	c ₁	d ₁		
a ₂	b ₂	c ₂	d ₂		
a ₃	b ₃	c ₁	d ₁		

- (1) Calculate the relational algebra operation: $\Pi_{A,C,D} (A \neq a_1 (R)) \div S$
- (2) Express the division operation using domain relational calculus.

4. Please answer the following questions:

With the coming of the World Cup, how to manage football activities has become a hot topic. In order to solve this problem, we built a Football Activity Management System, which needs to maintain the information of the Team, Player, Chief Coach, Chief Referee, Sponsor, And each Match. The data schema is set as follows:

- [1] Team (TeamId, TeamName, PlayersNum, EstablishDate, ChiefCoach) ;
- [2] Player (PlayerId, PlayerName, BirthDate, Height, Weight) ;
- [3] Chief Coach (CoachId, CoachName, BirthDate, CoachLevel) ;
- [4] Chief Referee (RefereeId, RefereeName, BirthDate, RefereeLevel) ;
- [5] Sponsor (SponsorId, SponsorName)
- [6] Match (MatchId, HomeTeamId, VisitTeamId, ChiefRefereeID, MatchDate, Score).

Assume that the football system meet the following arrangements:

- [1] Each team has only one chief coach and several players.
- [2] Each chief coach can only be hired by one team, and each player only play for one team.
- [3] There are two teams in each match, one for the home team and one for the visiting team. There is only one chief referee for each match. Each match has a unique id and the date and score of the match are recorded.
- [4] Each sponsor can sponsor multiple teams, but each team can only have one sponsor. Each sponsor can individually sponsor multiple players, and each player can also represent multiple sponsors.

[5] Additionally, it allows the cases where one team plays more than one matches in the same day or plays with the same team in the same day.

- (1) Please draw the ER diagram of the football activity management system according to the above Settings (5).
- (2) Please write Primary Key,Candidate Key and Foreign Key of the Match table(10).

5. Please answer the following questions:

The relationship between Teachers and Courses is shown in the following table:

Course ID(key)	Teacher Name	Department ID
C1	David	D1
C2	Mally	D2
C3	Musk	D1
C4	Musk	D1

- (1) Which paradigm does the above relationship belong to? Why? (5)
- (2) Does there exist the problem of "Delete Exception" or "Modify Exception"? If it does, how do they happen? (5)

6. Please answer the following questions:

It is known that the data relationship of local enterprises is as follows:

[1] Information of Employees: EMP(Eno, Ename, Sex, Age, Title)

[2] Information of Companys: CPY(Cno, Cname, Area)

Please answer the question according to the above information:

- (1) Please CREATE an information table WORKIN where employees work in the enterprise and meet the following conditions: (10)

[1] The WORKIN table contains three attribute (the Eno from Emp, the Cno from CPY and the Salary)

[2] Define the primary and foreign key.

[3] Set a constraint that the Salary should not be less than 3,000 yuan.

- (2) Based on the WORKIN table created in question (1), fill in the following SQL statement implementation function:

- a) Query the Cno and Cname of the company which has the most employees (2+3)

```
SELECT C.Cno,C.Cname
FROM Company AS C, WORKIN AS W
WHERE C.Cno = W.Cno
GROUP BY ____ (A)
HAVING ____ (B) (SELECT COUNT(*)
FROM WORKIN AS W2
GROUP BY W2.Cno)
```

- b) Query the Cno and Area of the Company whose employees have the oldest average Age (2+3)

```

SELECT C.Cno, C.Area
FROM EMP AS E, CPY AS C, WORKIN AS W
WHERE W.Cno = C.Cno
AND W.Eno = E.Eno
GROUP BY C.Cno, C.Area
HAVING ____ (C) ____ >= ____ (D) ____

```

- (3) Please fill in the vacancy part of the following **TRIGGER** to realize the automatic maintenance function of employee salary. When the Title of the employee changes, the salary of the new title should be calculated according to the "Title-Salary" Function——SalaryList(Title) . And the salary should be updated automatically: (1+2+2)

```

CREATE TRIGGER NewSalary
AFTER UPDATE ____ (E) ____ ON EMP
REFERENCING NEW AS N
FOR EACH ROW
BEGIN
    UPDATE WORKIN
    SET ____ (F) ____
    WHERE ____ (G) ____
END

```

7. Please answer the following questions:

Suppose there is a Electronic Toll Collection (ETC) related to the highway:

When a vehicle drives out of the station, firstly, we read the current balance X from its credit card A , and we records it as $\underline{X} = R(A)$; Then we deduct the cost S and we records it as $\underline{X} = X - S$; Finally, we write the balance X into credit card A , and we records it as $W(A, X)$. We also assume that more than one vehicles can use the same credit card at the same time.

Please answer the following questions based on the above assumptions:

- (1) If two vehicles $T1$ and $T2$ bound to the same credit card A drive out of the cost station at the same time, their order execution sequence is as follows:

$T1: X1 = R(A), X1 = X1 - S1, W(A, X1),$

$T2: X2 = R(A), X2 = X2 - S2, W(A, X2),$

What problems will occur at this time? Could you explain why? (5)

- (2) To solve the above problems, the exclusive Lock $XLock(A)$ is introduced to lock data A , and the UnLock $UnLock(A)$ is introduced to unlock data A . Please supplement the above execution sequence to make it meet the 2PL protocol. (5)

- (3) When two vehicles bound to the same credit card number pass through the toll booth at the same time, the possible sequence of instruction executions is: $x1=R(A)$, $x1 = x1 - a1$, $x2 = R(A)$, $x2 = x2 - a2$, $W(A, x1)$, $W(A, x2)$. To solve the above problem, the exclusive lock instruction $XLock(A)$ is introduced to lock the data A , and the unlock instruction $UnLock(A)$ is used to unlock the data A . Please supplement the above execution sequence to satisfy the 2PL protocol.(5)